Evaluation of open-source voice cloning models for ALS patient: A case study using legacy audio data

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ABSTRACT

Amyotrophic lateral sclerosis (ALS) deprives most patients of intelligible speech, often before a voice can be banked. This case study asks whether a voice can instead be reconstructed from legacy material. From 42 WhatsApp voice messages of a German-speaking patient, recorded before diagnosis, three zero-shot voice cloning models (Qwen3-TTS, Fish Audio S2 Pro, Voxtral TTS) were enrolled with a single cleaned reference and compared against the patient's own recordings as a fourth condition. Speaker similarity, word error rate, response time were complemented by a naturalness test and a voice Turing test with 30 raters. Two of the three models were rated as natural as the genuine recordings, while none passed as the patient in a direct comparison. The patient herself identified her own voice in every pair. Speaker similarity ranked the systems in reverse order of perceived naturalness: the system closest to the patient by embedding was rated least natural and detected most reliably. S2 Pro is recommended as the only system that meets all requirements. The results show that half an hour of everyday recordings suffices for an assistive voice, and that objective similarity alone is not a sufficient selection criterion.

Keywords: Voice Cloning, Text-to-Speech, ALS, Open Source, Low-Resource

INTRODUCTION

Communication is arguably the most fundamental human ability. One of its most personal carriers is the voice. A voice does more than transport words. It identifies the person speaking and most people can recognize someone they know without seeing them at all.

This is why the loss of speech in progressive neurological disease is not only the loss of a function. Retaining a voice that still sounds like one's own is described as strengthening personal identity in social situations, easing the fear of losing the ability to communicate and helping patients remain autonomous as their condition advances[16]. Accordingly, studies on "voice banking" in motor neuron diseases report that preserving one's identity, along with the desire to maintain existing social relationships, is the primary motivation for its use[10].

Amyotrophic lateral sclerosis (ALS) makes this loss the rule rather than the exception. Between 80 % and 95 % of people with ALS eventually reach a point at which their natural speech can no longer be understood[4]. Approximately 30 % of affected individuals initially exhibit bulbar symptoms, that is, symptoms originating in the brainstem, whose motor neurons control the muscles responsible for speech and swallowing, as reported in [6]. The majority, around two-thirds present with a spinal form of the disease, meaning that symptoms begin in the extremities [19].

While established approaches address this problem, they rely on conditions that cannot be met by patients suffering from a sudden onset of voice impairment or a diagnosis made too late. "Voice banking" captures a person's voice while it is still available, in order to build a personalized synthetic voice from it later[21]. Depending on the system, this process requires between 300 and 500 recorded sentences or phrases[10] and around 1800 utterances for unit-selection systems[21], a voice that is still sufficiently intelligible and the physical stamina to complete the process. However, time is often a critical factor working against the patient. As speech intelligibility declines, speaking ability frequently deteriorates so rapidly that there is little time left to implement appropriate measures[4]. Consequently, the voice is often already impaired by the time of diagnosis, meaning the necessary speech material is no longer

available in the required quality. Recording at that point doesn't solve the problem either. If speech is already impaired, speaker adaptation reproduces not only the speaker's identity but also the symptoms of the impairment, such that the synthetic voice itself is impaired[21].

Obtaining the assistive technology itself consumes further time the patient does not have. Such devices are referred to as augmentative and alternative communication (AAC) systems, which supplement or replace natural speech, ranging from letter boards to eye-controlled speech-generating devices. The health care systems of many countries finance high-tech AAC devices only partially or not at all[12]. Even where funding exists, the process is slow. On average, 93 days elapsed between the request and the provision of a device. When provision failed, the reason was usually a denial by the health insurance provider. Less commonly, patients themselves decided against the device or died before it arrived[12].

What most people do have, however, is legacy material: voice messages, videos or recordings made years before the diagnosis, for entirely unrelated reasons. The goal of this work is to determine whether such material is sufficient to create a usable synthetic voice for an ALS patient and which of three current zero-shot cloning models is best suited for that purpose.

CONCEPT AND IMPLEMENTATION

0.1 Data basis

In this study the data basis originates from a 56 year-old native german-speaking female person. She was diagnosed with ALS in 2024, although speech-related symptoms had already appeared in April 2022, leading to a rehabilitation stay in 2022. The patient suffers from an uncommon form of ALS in which the voice is lost first. A total of 42 voice messages were collected from WhatsApp, spanning the period from 2021 to 2022. The participant provided written consent for the use of her audio data in this study.

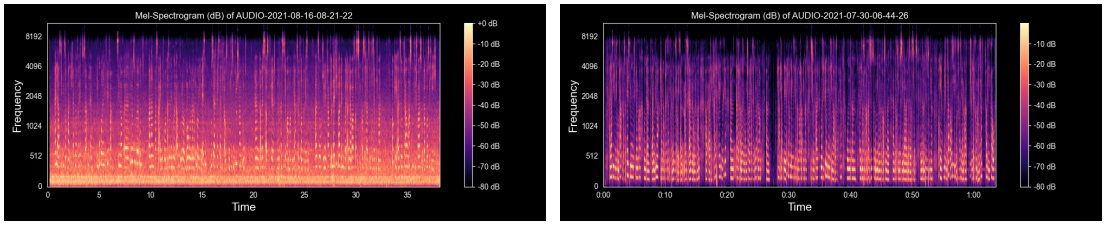
0.2 Exploratory Data Analysis

As a first step basic statistics were computed for the audio files. The exported files comprise a total duration of 1858.1 seconds, corresponding to 30.9 minutes (Table 1). Furthermore the duration statistics of the audio files were analyzed, as file length directly impacts the fine-tuning process. Large audio files can cause out-of-memory errors during the computation. The file splitting was required for the fine-tuning approach only and is not part of the final pipeline. The analysis revealed a minimum duration of 13.82 seconds and a maximum of 152.60 seconds, with a mean of 44.24 seconds and a standard deviation of 28.94 seconds. This indicates a considerable variance in file duration, with a coefficient of variation ($CV = \sigma/\mu$) of 65%.

Source	Amount of files	Total duration	Format	Sample rate
Voice Messages	42	1858,1 s	.m4a	22 050 Hz

Table 1. Overview to the dataset statistics

Since most recordings were not captured under professional conditions, background noise is expected to be present in parts of the dataset. To assess the extent of noise contamination, Mel-spectrograms were generated for each recording.



(a) Mel-spectogram analysis with background noises **(b)** Mel-Spectrogram analysis without background noises

Figure 1a depicts the Mel-spectrogram of a representative recording. A persistent horizontal band of high spectral energy is observed in the low-frequency range (0–512 Hz) across the entire duration of the

recording. Based on spectral analysis, this pattern can be attributed to constant background noise, which was confirmed through manual auditory inspection.

In contrast, Figure 1b displays a cleaner spectral distribution without significant background noise. Clearer vertical energy patterns are visible across the low to high-frequency range, corresponding to spoken segments. This comparison illustrates the variability in recording quality across the dataset.

Identifying and addressing such noise is critical, as unfiltered background noise may cause the model to learn irrelevant acoustic features rather than the patient's vocal characteristics.

0.3 Data Preprocessing

The first step of the audio data preprocessing is to convert the existing files into a suitable format for the modelling. That is why it is recommended to convert into .wav audio format and mono structure.

To address the background noise issue from the further analysis, DeepFilterNet2 is applied for speech enhancement.[17] This approach is a deep learning based speech enhancement tool and consists of two stages. The first stage is responsible for improving the speech envelope of the audio file, while the second stage applies deep filtering to refine the harmonic structure, which is relevant for the speech quality. The framework was selected due to low computational overhead and open source availability.[18] The tool effectively reduced the background noises observed in Figure 1a, as shown in Figure 2. A manual auditory inspection revealed no audible deterioration in speech quality after the enhancement.

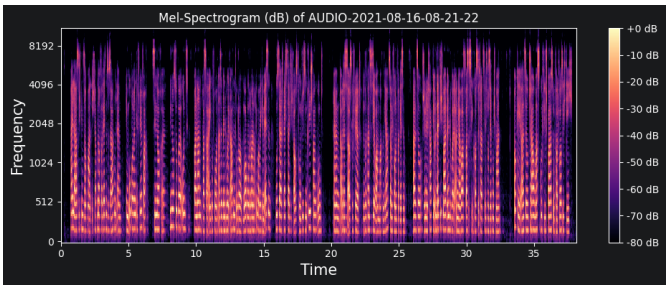


Figure 2. DeepFilterNet2 applied to Figure 1a without background noises

The messages were recorded over two years under different conditions like varying distances from the microphone and at volumes ranging from a louder voice to almost whispering. A way to approch this problem is by applying a high-pass filter at 80 Hz removes rumble from the phone microphone, a compressor with a threshold of -16 dB, reduces the dynamic range within the file. To ensure a consistency on all the files, LUFS (Loudness Units Full Scale) normalization is applied to the audios and a limiter at -1 dBFS prevents the applied gain from clipping. LUFS is defined in ITU-R BS.1770[9] and estimates *perceived* loudness. It weights frequencies the way human hearing does and ignores pauses. An RMS (Root Mean Square) value would instead average over the pauses and a peak value would depend on single sample such as a microphone bump. The recommended LUFS value that is also beeing used for the preprocessing is -24[3].

Pauses are removed in a second step. The threshold is derived from the data rather. Therefore the frame-wise RMS energy of every file is converted to decibels and the noise floor and the speech level are estimated as the 10th and the 80th percentile. Percentiles are used instead of minimum and maximum because a single outlier frame would otherwise define the range.

0.4 Model choice

Before the implementation it is important to define some requirements that the models should fulfill. First of all it would be important that the model is able to process and speak in german. It should also be open source, to have the possibility to host the model on a private server, because of privacy reasons. Regarding the inference the models ought to provide low latency or streaming support for a later implementation in an app. This point is very important for a normal usage because a high latency would impede the communication with the app drastically. Concerning the minimal data the models should provide fine-tuning or zero-shot learning. While fine-tuning requires a larger amount of data, zero-shot learning needs significantly less data.

0.4.1 Qwen3-TTS

Qwen3-TTS(text-to-speech) is a multilingual text-to-speech model that supports ten languages, including German. It enables adaptive control of tone, speaking rate and emotional expression based on additional instructions. The model family is released in two sizes, whereas the implementation uses the 1.7B parameter variant. The Qwen team report a very low streaming latency, which they attribute to a dual-track autoregressive architecture that allows decoding to start from the first frame, yielding a first packet latency of 101 ms. An efficient speech tokenizer compresses the audio into a compact representation while preserving prosodic an emotional nuance. The base variant of the models is able to create a voice clone from a 3-second reference example and is also good for fine-tuning.[8]

0.4.2 S2 Pro

Fish Audio S2 combines a dual-autoregressive architecture with reinforcement learning alignment. The speech generation is split along two axes. On the one hand a Qwen3-4B backbone (Slow AR) advances one step per frame and predicts only the semantic codebook, while the lightweight four-layer decoder (Fast AR) generates the nine remaining acoustic codebooks. The model renders emotional expressions by using free-form textual descriptions such as whisper, angry and can be injected at specific word positions. The model was pre-trained on over ten million hours of audio and including 80 languages and dialects. The model supports a real-time factor of 0.195 and a time-to-first audio of approximately 100 ms.[11]

0.4.3 Voxtral 4B TTS

Voxtral TTS is a multilingual text-to-speech model supporting nine languages, including German, which generates speech from a reference recording as short as three seconds. The team behind Voxtral designed the architecture differently from that of the other models. Voxtral and the models discussed above break down the language into a semantic and an acoustic layer. While this break down is performed autoregressively in Qwen3 and S2 Pro is doing autoregressive, Voxtral takes a hybrid approach. The semantic tokens are generated autoregressively in a decoder-only backbone, while the acoustic tokens are generated using a flow-matching transformer. According to the report, flow matching is more efficient than the autoregressiv approach in terms of quality, computational cost and latency. The model provides German support and is also capable of creating a voice clone based on a three seconds reference recording. Furthermore it doesn't support fine-tuning. The development team reports a latency of 70 ms for the first audio chunk, making it suitable for real-time applications.[13]

Since the application is intended for potential deployment, the non-commercial licence of Voxtral TTS precludes it as a deployment candidate. It is nevertheless retained as a baseline with a 68.4 % win rate over ElevenLabs Flash v2.5 in human evaluations of multilingual zero-shot voice cloning, it provides a competitive reference against which the two remaining candidates can be assessed.[13]

0.5 Tech stack

The pipeline is implemented entirely in Python. Audio preprocessing utilizes DeepFilterNet2 for speech enhancement, alongside Pedalboard combined with librosa for filtering and loudness normalization. Access to the three synthesis models is provided via the supplied SDK or a WebSocket client. PyTorch is used to calculate objective metrics. SpeechBrain provides the ECAPA-TDNN speaker embeddings, while OpenAI Whisper is used for transcriptions, the quality of which is evaluated using jiwer. To simplify the analysis of the listening test, a self-hosted web application with a lightweight browser frontend was developed. NumPy, pandas and statsmodels are used for analyzing the results. The source code and results are available in the project repository.[1]

0.6 System architecture

The first approach to create a voice clone was to use fine-tuning. On the one hand fine-tuning generated a good voice cloning model, but the fine-tuning took too long and used too much computational power. In addition, zero-shot learning was evaluated. The output audio was of comparable quality to that of the fine-tuned model, while requiring fewer resources and less computational time. It should be mentioned that this comparison rested on a listening test rather than objective metrics. Therefore zero-shot learning was the preferred way for the implementation. Another argument for zero-shot learning is the minimal amount of data. If a person is already suffering from a speech disorder or has had a sudden stroke, it is not

[1] https://github.com/seiferla/als-voice-cloning-evaluation

possible to generate a large amount of audio data. This is precisely where zero-shot learning demonstrates its strength compared to fine-tuning.

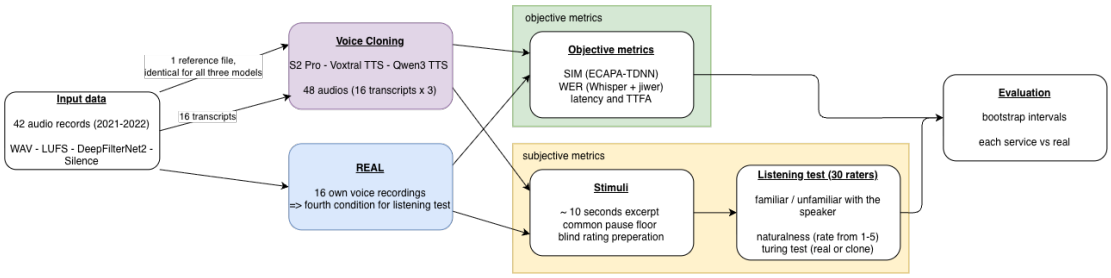


Figure 3. Overview of the evaluation pipeline: the three cloning services and the genuine recordings (REAL) render the same texts and are compared through parallel objective and subjective branches.

EVALUATION

The evaluation of the voice cloning systems is structured into three complementary parts. First, objective metrics are computed automatically and provide a reproducible baseline for speaker similarity, intelligibility and latency. Second, a subjective listening test (N-MOS) captures human perception of the naturalness of a selected subset from the audio files. This subset also contains genuine recordings, to detect whether the raters are able to identify the real audios as human or not. Third, a voice Turing test assesses whether listeners are able to distinguish the cloned voice from the original recordings of the patient. All three parts rely on a shared test set described in Section 0.7.

0.7 Test Set

To enable a fair comparison across the three evaluated models (Qwen3, Fish Speech S2 Pro and Voxtral Mistral), a common test set of $N = 16$ German sentences was defined. The transcripts were produced with Whisper and corrected by hand, giving each sentence a known ground-truth text, which is important for the word error rate metrics (Section 0.8). These records vary naturally in length of 12-92 seconds.

All test sentences were synthesized in a zero-shot voice cloning via the provided APIs of the models, using the same reference sample of the patient's voice for all three systems to ensure comparability. The patient's own recordings of the same 16 sentences were carried through the evaluation as fourth, equally sized condition, rated alongside the three cloned systems in the listening test and used as the original class in the Turing test. To guarantee a valid evaluation, different evaluation plans were introduced, where each rater was randomly assigned one of six pre-built plans. Each plan split that rater's 13 texts into two disjoint halves, one scored for naturalness and one used in the Turing pairs. This ensures that no rater saw the same text in both tasks. Rotating which model filled which role across plans ensured every text-model combination was rated an equal number of times over the full sample.

A total of $n = 30$ participants took part in the evaluation, where all of them were native German speakers. The participants were divided into two groups. One group know the person's voice or had seen them before and the other group had never seen them or heard their voice.

0.8 Objective Metrics

The objective evaluation quantifies speaker similarity, intelligibility and API performance without requiring human raters.

Speaker Similarity (SIM) is computed as the cosine similarity between speaker embeddings extracted with ECAPA-TDNN [5] from the synthesized samples and from original audio reference of the patient, using the SpeechBrain toolkit [15]. Higher values indicate a closer match to the target voice.[7]

Word Error Rate (WER) is obtained by transcribing each synthesized sample with Whisper [14] (`small`) and comparing the transcription against the ground-truth text of the corresponding test sentence. A high WER indicates unclear articulation.[7]

Latency and Real-Time Factor (RTF) are derived from the end-to-end response time of each API call. RTF is computed as the ratio of generation time to the duration of the generated audio [1]. A lower RTF indicates a faster synthesis relative to audio duration and is therefore preferable for real-time applications.

During the synthesis phase, each text was generated three times per model and analyzed in the format provided by the service, without further processing. Results for each text were averaged across the three repetitions. The median served as the statistical summary across all texts. The 95 % confidence interval was obtained from 4,000 bootstrap replicates, in which texts were resampled with replacement and the median recomputed each time. The interval is given by the 2.5th and 97.5th percentiles of the resulting distribution.

Differences between conditions were computed per text, so that both conditions are evaluated on the same sentences. They were judged at a Bonferroni-adjusted level, since four conditions give six comparisons per metric and the family-wise error rate would otherwise reach approximately 26 %. A difference whose adjusted interval contains zero is reported as *not detectable*, which is a statement about the resolution of the measurement, not about equality.

While Qwen3 and S2 Pro generated all 16 texts with usable quality, three Voxtral renders exhibited no measurable pause floor. Following the rule fixed in advance, the affected texts were excluded from the listening test for all conditions (Section 0.9).

Condition	**SIM** ↑	**WER** ↓	**RTF** ↓
Qwen3	0.755 [0.738, 0.769]	0.097 [0.075, 0.115]	0.392 [0.377, 0.410]
S2 Pro	0.703 [0.687, 0.713]	0.119 [0.101, 0.142]	0.227 [0.216, 0.249]
Voxtral	0.609 [0.602, 0.627]	0.116 [0.107, 0.141]	0.143 [0.107, 0.158]
Real	0.786 [0.760, 0.825]	0.190 [0.147, 0.228]	—

Table 2. Objective results. Median across texts with bootstrap 95 % confidence intervals; $n = 16$ texts per system, $n = 15$ for the reference condition, which excludes the recording used as the similarity reference. RTF applies to the three services only.

Table 2 illustrates the results of the objective evaluation. In similarity the systems rank Qwen3, S2 Pro, Voxtral and all differences between systems are detectable. The difference between Qwen3 and the real recordings (-0.031, adjusted CI $[-0.068, +0.004]$) is not detectable, whereas S2 Pro and Voxtral remain measurably below the reference. This describes the resolution of ECAPA-TDNN on this material, not a faithful reproduction of the patient's voice.

All three systems reach a lower WER than the genuine recordings (0.097 to 0.119 against 0.190). This is not a gain in intelligibility: the systems read a prepared transcript, whereas REAL is spontaneous phone speech. The difference between Qwen3 and S2 Pro is not detectable (-0.029, adjusted CI $[-0.061, +0.001]$).

Response time ranks in the reverse order. While Voxtral is the fastest (service RTF 0.143), is Qwen3 the slowest (0.392) of the evaluated models, with all pairwise differences detectable. All three stay below 1.0 and thus generate audio faster than it is spoken.

0.9 Subjective Evaluation (N-MOS)

To consider aspects of speech quality that are not reliably reflected in automatic metrics, like naturalness and prosody, a Naturalness Mean Opinion Score (N-MOS) listening test was conducted. Participants listened to individual synthesized samples, without any reference to the original recording and rated the naturalness of each sample on a 5-point scale (1 = clearly unnatural, 5 = fully natural, sounds like a human).[20] Genuine recordings of the patient were included at random positions as hidden anchors to analyze whether the participants were still able to recognize a real voice and classify it as such.

Stimuli are excerpts of about 10 s (5.9 to 14.4 s), cut on sentence boundaries so that every condition presents the same passage of a text. All stimuli, including the genuine recordings, passed through identical processing: resampling to 24 kHz, loudness normalisation to -23 LUFS [9] and a common pause noise floor. The last step is required because the denoised recordings otherwise have digitally silent pauses that no synthesis system produces.

The score of a condition is the mean of per-rater means, so that every rater carries equal weight. Intervals come from a two-way cluster bootstrap over raters and texts, since answers from one rater and answers about one text are each correlated. Differences are formed within rater, which removes each rater's scale offset and judged at a Bonferroni-adjusted level over six comparisons. A simulation of this

design before data collection detects a difference of 0.30 MOS with 80 % power at 30 raters, so smaller differences are reported as not detectable rather than as absent.

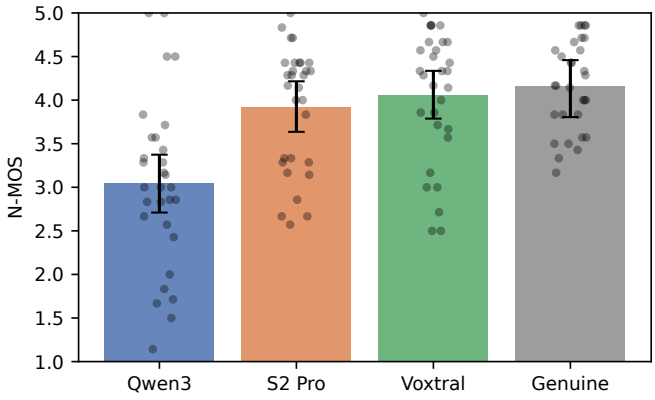


Figure 4. Naturalness by condition. Bars are the mean of per-rater means with two-way cluster bootstrap 95 % intervals (30 raters, 13 texts). Each dot represents the rater's mean for that condition.

Figure 4 depicts the result. Qwen3 scores 3.04 $[2.71, 3.37]$ and lies 0.9 to 1.1 points below every other condition. S2 Pro (3.92 $[3.64, 4.22]$), Voxtral (4.05 $[3.79, 4.34]$) and the genuine recordings (4.16 $[3.81, 4.46]$) are not distinguishable from each other: none of the three adjusted intervals excludes zero. The hidden repeats set the scale for this. Across 60 pairs, a rater rating the same stimulus twice differed by 0.55 points on average, which is more than the gaps among these three conditions (0.10 to 0.23) and far less than the Qwen3 gap.

The ranking is the reverse of the similarity ranking in Table 2. The system whose clone is closest to the patient by the speaker embedding is the one raters find least natural and the model with the lowest similarity is not distinguishable from the genuine recordings. The preregistered hypothesis that naturalness follows similarity is therefore rejected and the finding is taken up in Section 0.11.

0.10 Voice Turing Test

As a final and more stringent test of perceptual quality, a voice Turing test was conducted to assess whether listeners can reliably distinguish cloned samples from original recordings of the patient. In each trial, the participants heard two blinded excerpts of the same sentence, one of which was an audio recording of the patient's own voice and the other a cloned version. The raters should identify which recording was the authentic one. The fooling rate is defined as the proportion of trials in which the cloned sample was misidentified as original. A fooling rate of 0.5 corresponds to chance level, meaning raters couldn't reliably distinguish the clone from the real voice, while a value below 0.5 indicates that the clone was detected more often than chance.[20]

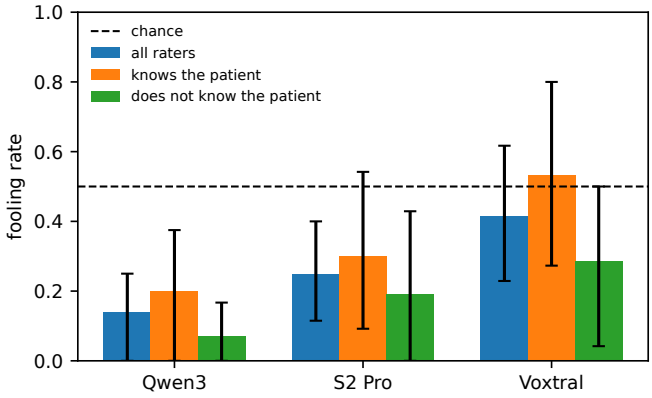


Figure 5. Fooling rate by model and rater group with two-way cluster bootstrap 95 % intervals. The dashed line is chance level.

Detection differed sharply between systems (Figure 5). Qwen3 was identified most reliably at a fooling rate of 0.14 $[0.00, 0.25]$, followed by S2 Pro at 0.25 $[0.12, 0.40]$; both intervals lie below chance. Voxtral reached 0.41 $[0.23, 0.62]$, and its interval includes 0.50 in every rater group, so detection is not established for it on this material. The ranking matches the naturalness ranking.

An interesting finding is that raters who know the patient were fooled more often than strangers, for all three systems. The effect is largest for Voxtral, where this group chose the clone in 0.53 $[0.27, 0.80]$ of pairs. All three contrasts ($+0.13$, $+0.11$, $+0.25$) have intervals containing zero, so at 15 versus 14 raters this is a trend, not a result. The patient herself, rated in a separate session and kept out of all averages, identified her own voice in 6 out of 6 cases.

0.11 Model Selection

In conclusion, the S2 Pro is the most suitable of the evaluated models for the intended application. This decision is based on a process of elimination. Qwen3 is ruled out due to its perceived quality. Despite offering the highest speaker similarity, it is rated as significantly less natural than all other variants and is the most easily identified in the Turing test. Based on the figures, Voxtral is on par with the S2 Pro, but is excluded due to its non-commercial license and the fact that it was intended to serve as a reference baseline. The recommendation is not based on S2 Pro sounding better than Voxtral, but rather on the fact that it is the only system that meets all requirements simultaneously (Table 3). Whether the recommended system can be self-hosted, as the requirements demand, was not tested here and has to be established before deployment.

	Qwen3	**S2 Pro**	**Voxtral**
Naturalness vs. genuine	below	not distinguishable	not distinguishable
Fooling rate	0.14	0.25	0.41
Usable renders	16/16	16/16	13/16
RTF	0.39	0.23	0.14
Deployable licence	yes	yes	no

Table 3. Selection criteria. A system is recommended only if it passes every row.

DISCUSSION

The clearest result is a disagreement between the two kinds of measurement. On the one hand Qwen3 is closest to the patient by speaker embedding, so close that ECAPA-TDNN cannot separate it from a second genuine recording and the other hand it is the system raters find least natural and detect most reliably. Voxtral is the mirror image (Figure 6). Both perceptual measures agree with each other and contradict the objective one. A selection by speaker similarity alone would have picked the worst system for this purpose.

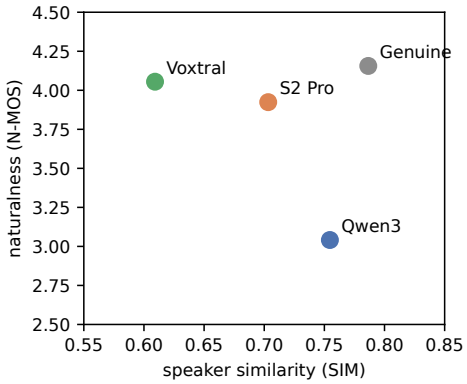


Figure 6. Speaker similarity against naturalness. The system closest to the patient by embedding is the one raters find least natural.

The disagreement concerns the similarity, not the objective measurement itself. Listeners cannot perceive reaction times, yet these represent a practical limitation. Although every model operates faster than real-time, the median end-to-end latency ranged from 5.7 to 13.8 seconds per utterance due to the length of the texts. A dialogue oriented application would need to perform synthesis in shorter segments or synthesize the text while the user is still typing.

The reversed familiarity effect cannot be interpreted. No unfamiliar rater was older than 42, the familiar group has a median age of 51 and an age-related loss of high-frequency hearing would blunt exactly the brightness cue that separates genuine from cloned excerpts; hearing was not screened. Beyond this, the study covers one speaker, the fooling rates are a lower bound because pause structure, brightness and duration remain usable cues, and the surviving texts are those Voxtral rendered, which flatters it.

CONCLUSION

This study asked how much of a voice can be reconstructed from material that was never recorded for that purpose. Three zero-shot models were compared against the patient's own recordings as a fourth condition, using objective metrics, a naturalness test and a voice Turing test with 30 raters. Two of the three systems are rated as natural as the genuine recordings, none passes as the patient in a direct comparison and S2 Pro is recommended for further experiments or implementation (Section 0.11). Legacy material is therefore sufficient for an assistive voice, if not for a replica of the original.

Several insights can be drawn from this work. First, zero-shot cloning based on a reference recording lasting only a few seconds performed just as well in listening tests as the fine-tuned model and at a fraction of the computational cost, thereby rendering fine-tuning unnecessary. Second, this change did not make preprocessing superfluous, as had initially been assumed. DeepFilterNet removed background noise from the phone recordings without leaving any audible artifacts and all three services were enrolled with this cleaned reference. Third, objective measurements pointed to the wrong system, which was only revealed in the listening test.

ACKNOWLEDGMENTS

This work was largely written independently. Claude contributed additions to the methodology documentation and the model evaluation and implemented the evaluation framework and its automation based on the student's concepts. Claude Code is also used for the documentation in the repository. Further all contributions were reviewed and verified by the student [2].

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